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SARDAR PATEL UNIVERSITY
M.Sc.(Information Technology)
SEMESTER-I

External Examination

PS01CINT21- Introduction to Theoretical Computer Science

16th Nov, 2019, Saturday

Time:10:00am to 1.00pm

Marks:70

Q1

Give answers of following Multiple choice questions

[8]

- 1 $a \wedge a = a$ is _____.
A Commutative property B Absorption property
C Idempotent property D Associative property
- 2 In lattice, ____ is used to denote universal lower bound and ____ is used to denote universal upper bound.
A 0,0 B 0,1
C 1,0 D 1,1
- 3 The set $X = \{1, 2, 3, 4\}$ is an example of _____ set.
A Infinite B Uncountable and Infinite
C Finite D Uncountable
- 4 Replacing join operation over meet and meet operation over join is known as principle of _____.
A Replacement B Duality
C Durability D None of these
- 5 A graph that has neither self-loops nor parallel edges is called a _____ graph.
A Simple graph B Multi Graph
C Null graph D None of these
- 6 The fire in a factory is an example of _____.
A Secular Trend B Irregular Variation
C Seasonal Variation D Cyclical Variation
- 7 Increase in the number of patients in the hospital due to heat stroke is _____.
A Secular Trend B Irregular Variation
C Seasonal Variation D Cyclical Variation
- 8 Measuring the time, takes to execute an algorithm is known as _____.
A Time Complexity B Cost Complexity
C Mapping D None of these

Q2

Do as directed (Any 7)

[14]

- 1 Write down the equation for straight line and parabolic form.
- 2 Write a short note on tractable and intractable problems.
- 3 What is algorithm? List the characteristics of good algorithm.
- 4 Construct 5-yearly moving averages of the number of students studying in a college shown below:

①

(P.T.O.)

Year	No. of Students	Year	No. of Students
1998	332	2003	405
1999	317	2004	410
2000	357	2005	427
2001	392	2006	405
2002	402	2007	438

- 5 Define Complete Graph. Draw Complete Graph K_3 , K_4 and K_5 .
- 6 Construct a Circuit diagram for: (1) $XY + YZ(Y+Z)$ (2) $YZ + X(YZ' + Y'Z)$
- 7 Explain finite and infinite sets using with proper example.
- 8 Explain power set and cardinality of a set using with proper example.
- 9 Explain with example Hamiltonian Path and Hamiltonian Circuit.

Q3 [A] Define Relations. Explain and give example for each of the following. [6]
a) Reflexive Relation b) Antisymmetric Relation c) Equivalence Relation

Q3 [B] Discuss basic properties of lattices. Explain meet and join operation. [6]
Assume a partial order set $A = \{(2,4,6,8), /\}$. Then investigate the following:
a) Every pair of elements in the poset has GLB.
b) Every pair of elements in the poset has an LUB.
For true or false.

OR

Q3 [B] Define Lattice. List the types of lattice. Explain Distributive lattice with example. [6]

Q4 [A] Define POSET. Explain properties of POSETS. Draw the Hasse Diagram for the POSET [6]
 $\langle S_{60}, / \rangle$ and $\langle S_{24}, / \rangle$. Where the relation division (a/b) means a divides b.

Q4 [B] Write down the steps to find shortest path in a weighted graph. [6]

OR

Q4 [B] Explain the following term with proper example. [6]
a) Subgraphs b) Euler graphs c) Multigraphs

Q5 [A] Define graph. Discuss any two applications of graph in detail. [6]

Q5 [B] Fit a straight line trend by the method of least squares to the following data. Assuming [6]
the same rate of change continues what would be the predicted series of the year 2011?

Year	Sales (Rs. Lakhs)	Year	Sales (Rs. Lakhs)
2002	76	2006	138
2003	80	2007	120
2004	130	2008	174
2005	144	2009	190

Calculate trend values from 2002 to 2009.

OR

Q5 [B] Write down the steps for LARGEST1 and LARGEST2 algorithm. [6]



Q6 [A] Fit a straight line trend for the following data and estimate the value for the year 1960. [6]

Year	1950	1951	1952	1953	1954	1955	1956
Value ('000)	346	411	392	512	626	640	611

Q6 [B] Fit a 2nd degree parabola for the following data: [6]

Year	1996	1997	1998	1999	2000
Sales ('000)	40	50	62	58	60

OR

Q6 [B] What is Time-Series? List and explain the utility and components of the time series. [6]

— X —
③

